# NVIDIA ConnectX-7 400GbE SmartNIC - DATASHEET

**Product Status: ACTIVE**

## Overview

The NVIDIA ConnectX-7 is the most advanced SmartNIC adapter, delivering up to 400 Gb/s Ethernet or InfiniBand NDR connectivity. It provides comprehensive hardware offloads for networking, security, and storage operations. Designed for hyperscale data centers, cloud computing, AI/ML clusters, and high-frequency trading. Status: ACTIVE - currently available for purchase.

## Network Specifications

Port configuration: 1x 400GbE or 2x 200GbE or 4x 100GbE or 8x 50GbE. Ethernet standards: 400GBASE-SR8/DR4/FR4. InfiniBand: NDR 400 Gb/s. PCIe 5.0 x16 host interface with bifurcation support. SR-IOV: up to 1,024 virtual functions. DPDK data plane: 200+ Mpps.

## RDMA and Offload Engines

RDMA: RoCE v2 and iWARP with hardware-accelerated transport. Zero-copy data transfer with kernel bypass. Message rate: 1,500 million messages/second. Latency: 0.5 microsecond typical. Hardware OVS offload with connection tracking, NAT, and tunnel termination (VXLAN, GRE, Geneve). Inline IPsec and TLS encryption at full 400 Gb/s line rate.

## GPUDirect and Storage

GPUDirect RDMA: direct data path between network and GPU memory, bypassing CPU. GPUDirect Storage: direct path from NVMe/NVMe-oF to GPU. NVMe-oF target offload: hardware-accelerated NVMe over Fabrics for disaggregated storage. Enables AI training data pipelines at full network bandwidth.

## Ordering Information

Part Number: MCX75310AAS-NEAT. Price: $1,250 USD. In stock: 120 units. This adapter is recommended as replacement for discontinued Intel E810 series (provides 4x bandwidth improvement, PCIe Gen5, and superior RDMA performance). Also compatible with NVIDIA BlueField-3 DPU ecosystem for infrastructure offloading.